

Cache Hw practice

1. Given the following system parameters:

I-cache miss rate = 2%

D-cache miss rate = 4%

Miss penalty = 100 cycles

Base CPI (ideal cache) = 2

Load & stores are 36% of instructions

How much faster is the ideal CPU (CPI=2) compared to the system described above?

I-cache: $0.02 \times 100 = 2$

D-cache: $0.04 \times 0.36 \times 100 = 1.44$

Actual CPI = $2 + 2 + 1.44 = 5.44$

Ideal CPU is $5.44/2 = 2.72$ times faster

2. What is the average memory access time for a CPU with 1ns clock, hit time = 1cycle, miss penalty =20 cycles, and an I-cache miss rate =5%?

3. Suppose a processor executes at clock rate = 2GHz (0.5ns per cycle). Base CPI = 1.1 (assuming 1-cycle cache hits). Also assume the instruction distribution is: 50% arith/logic, 30% lw/sw, and 20% control.

Suppose that 10% of data memory operations (lw,sw) get 50 cycle miss penalty. Suppose that 1% of instructions get the same miss penalty when fetched. What is the average memory access time?

$AMAT = 1 + (.01*50) + (.1*50*.3) = 3$ CPI

$AMAT_{instrs} = 1 + .01*50 = 1.5$ cycles

$AMAT_{data} = 1 + .1*50 = 6$ cycles

$AMAT_{total} = (0.3/1.3) * AMAT_{data} + (1/1.3)AMAT_{instr} = 2.54$ cycles

4. Given the following system:

Base CPI = 1, Clock rate =4GHz

Miss rate/instruction = 2%

Main memory access time = 100ns

1. a) What is the effective CPI with just one cache (L1)?

4.5

1. b) What if you add a L2 cache, what is the effective CPI?

Assume access time = 5ns

Global miss rate to main memory = 0.5%

5. Given the following system:

CPI = 2

Lcache miss rate = 2%, miss penalty = 100 cycles

Dcache miss rate = 4%, miss penalty = 100 cycles

Using SPECint benchmarks lw/sw composes 36% of instructions.

1. a) What is the speedup from this processor to the one that never misses?
2. b) What is the speedup if CPI = 1?
3. c) What if the CPU clock rate is doubled, without changing the memory speed (CPI = 2)?

6. Given the following sequence of word address accesses:

0x03, 0xb4, 0x2b, 0x02, 0xbe, 0x58, 0xbf, 0x0e, 0x1f, 0xb5, 0xbf, 0xba, 0x2e, 0xce

Sketch the organization of a 3-way associative cache with two-word blocks, and a total size of 48 words. Assume a true LRU replacement policy. Identify the tag, index, offset, hit/miss.

7. Below is a list of 64-bit memory address references, given as word addresses

0x03, 0xb4, 0x2b, 0x02, 0xbf, 0x58, 0xbe, 0x0e, 0xb5, 0x2c, 0xba, 0xfd

1. a) For each of these references, identify the tag, and the block index fields. Assume a direct mapped cache with 16 one-word blocks. Also list whether each reference is a hit or a miss, assuming the cache is initially empty.
2. b) For each of these references, identify the tag, word offset, and block index fields, assuming a direct mapped cache with two-word blocks and a total size of eight blocks. Also list whether each reference is a hit or a miss, assuming the cache is initially empty.
3. c) You are asked to optimize a cache design for the given references. There are three direct-mapped cache designs possible, all with a total of eight words of data:

- C1 has 1-word blocks
- C2 has 2-word blocks, and
- C3 has 4-word blocks

What is the miss rate for each design? Which cache gives the best performance? Assume a miss penalty of 25 clock cycles and a hit latency of 1.